

Chapter 13: Data Storage Structures

1 Employee File (Figure 13.3)

Record	ID	Name	Department	Salary
0	10101	Srinivasan	Comp. Sci.	65000
1	12121	Wu	Finance	90000
2	15151	Mozart	Music	40000
3	98345	Kim	Elec. Eng.	80000
4	32343	El Said	History	60000
5	33456	Gold	Physics	87000
6	45565	Katz	Comp. Sci.	75000
7	58583	Califieri	History	62000
8	76543	Singh	Finance	80000
9	76766	Crick	Biology	72000
10	83821	Brandt	Comp. Sci.	92000

This table represents a fixed-length employee record file, where each entry is identified by its record number.

2 Q&A Discussion

2.1 Deletion Strategies

Question

Consider the deletion of record 5 (Gold, Physics, 87000). Compare the relative merits of the following techniques:

- (a) Move record 6 to the space occupied by record 5, and move record 7 to the space occupied by record 6, and so on.
- (b) Move record 7 to the space occupied by record 5.

(c) Mark record 5 as deleted, and move no records.

Answer

(a) Move record 6, then 7, etc.

- **Advantage:** Maintains compact storage and physical order.
- **Disadvantage:** Expensive if many records must be shifted; inefficient for large files.

(b) Move record 7 into record 5's space

- **Advantage:** Requires only one move.
- **Disadvantage:** Breaks logical ordering, making sequential scans less consistent.

(c) Mark record 5 as deleted (no movement)

- **Advantage:** Very efficient—no shifting required.
- **Disadvantage:** Leaves unused space unless a free-list reuse mechanism is implemented.
- **Usage:** Common in DBMS—deleted slots are later reused for inserts.

Conclusion: - Method (a): Suitable for small, ordered files. - Method (b): Minimizes moves but disrupts order. - Method (c): Most efficient; widely adopted with free-space management.

2.2 File Updates with Free List (Figure 13.4)

Question

Show the structure of the file of Figure 13.4 after each of the following steps:

- (a) Insert (24556, Turnamian, Finance, 98000).
- (b) Delete record 2.
- (c) Insert (34556, Thompson, Music, 67000).

header				
record 0	10101	Srinivasan	Comp. Sci.	65000
record 1				
record 2	15151	Mozart	Music	40000
record 3	22222	Einstein	Physics	95000
record 4				
record 5	33456	Gold	Physics	87000
record 6				
record 7	58583	Califieri	History	62000
record 8	76543	Singh	Finance	80000
record 9	76766	Crick	Biology	72000
record 10	83821	Brandt	Comp. Sci.	92000
record 11	98345	Kim	Elec. Eng.	80000

Initial structure of the file (Figure 13.4).

Answer

We use “ $\uparrow i$ ” to denote a pointer to record i .

(a) Insert Turnamian

header				$\uparrow 4$
record 0	10101	Srinivasan	Comp. Sci.	65000
record 1	24556	Turnamian	Finance	98000
record 2	15151	Mozart	Music	40000
record 3	22222	Einstein	Physics	95000
record 4				$\uparrow 6$
record 5	33456	Gold	Physics	87000
record 6				
record 7	58583	Califieri	History	62000
record 8	76543	Singh	Finance	80000
record 9	76766	Crick	Biology	72000
record 10	83821	Brandt	Comp. Sci.	92000
record 11	98345	Kim	Elec. Eng.	80000

Figure 13.101 The file after insert (24556, Turnamian, Finance, 98000).

Figure 1: After inserting Turnamian.

(b) Delete record 2

header				↑ 2
record 0	10101	Srinivasan	Comp. Sci.	65000
record 1	24556	Turnamian	Finance	98000
record 2				↑ 4
record 3	22222	Einstein	Physics	95000
record 4				↑ 6
record 5	33456	Gold	Physics	87000
record 6				
record 7	58583	Califieri	History	62000
record 8	76543	Singh	Finance	80000
record 9	76766	Crick	Biology	72000
record 10	83821	Brandt	Comp. Sci.	92000
record 11	98345	Kim	Elec. Eng.	80000

Figure 13.102 The file after delete record 2.

Figure 2: After deleting record 2.

(c) Insert Thompson

header				↑ 4
record 0	10101	Srinivasan	Comp. Sci.	65000
record 1	24556	Turnamian	Finance	98000
record 2	34556	Thompson	Music	67000
record 3	22222	Einstein	Physics	95000
record 4				↑ 6
record 5	33456	Gold	Physics	87000
record 6				
record 7	58583	Califieri	History	62000
record 8	76543	Singh	Finance	80000
record 9	76766	Crick	Biology	72000
record 10	83821	Brandt	Comp. Sci.	92000
record 11	98345	Kim	Elec. Eng.	80000

Figure 13.103 The file after insert (34556, Thompson, Music, 67000).

Figure 3: After inserting Thompson.

3 Future Questions (Template)

Question

[Write your question here]

For example: Explain how variable-length records are managed in database storage systems.

Answer

[Write your answer here]

You can use:

- Bullet points for pros/cons.
- Numbered lists for steps.
- Figures or tables if needed.